MIL-B-81757/13A(AS)
24 April 1987
SUPERSEDING
MIL-B-81757/13(AS)
28 February 1985

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM 24-VOLT, 30 AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of MIL-B-81757.

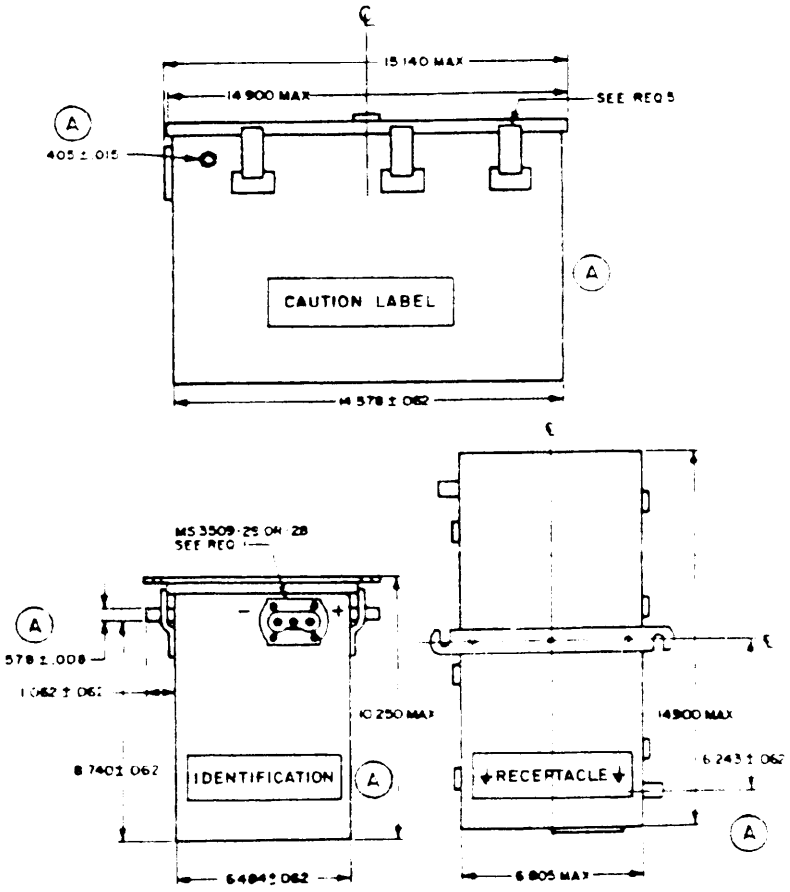


FIGURE 1. Battery assembly.
P/N M81757/13-1

AMSC N/A
FSC 6140
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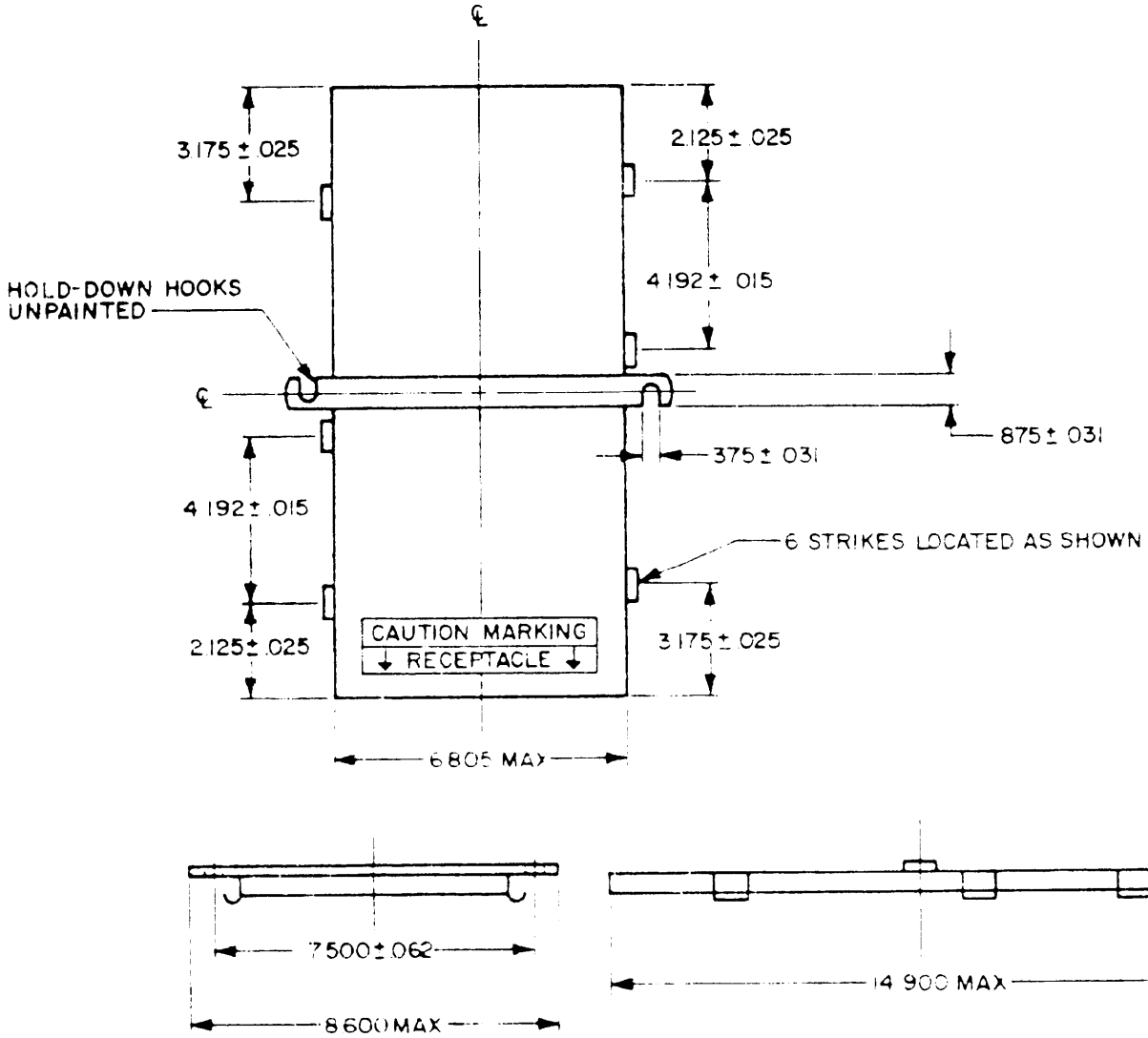


FIGURE 2. Cover.

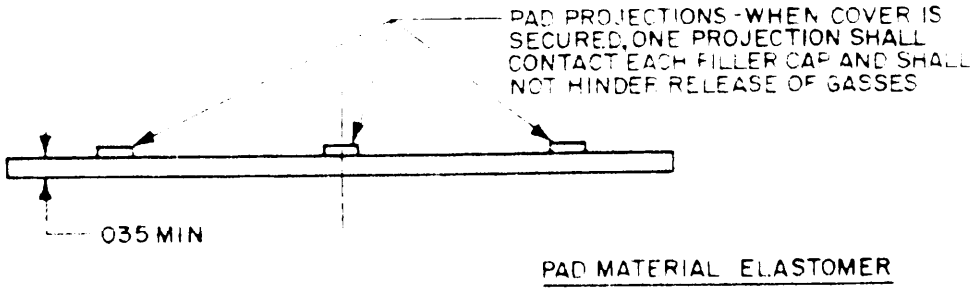


FIGURE 3. Cover hold-down pad.

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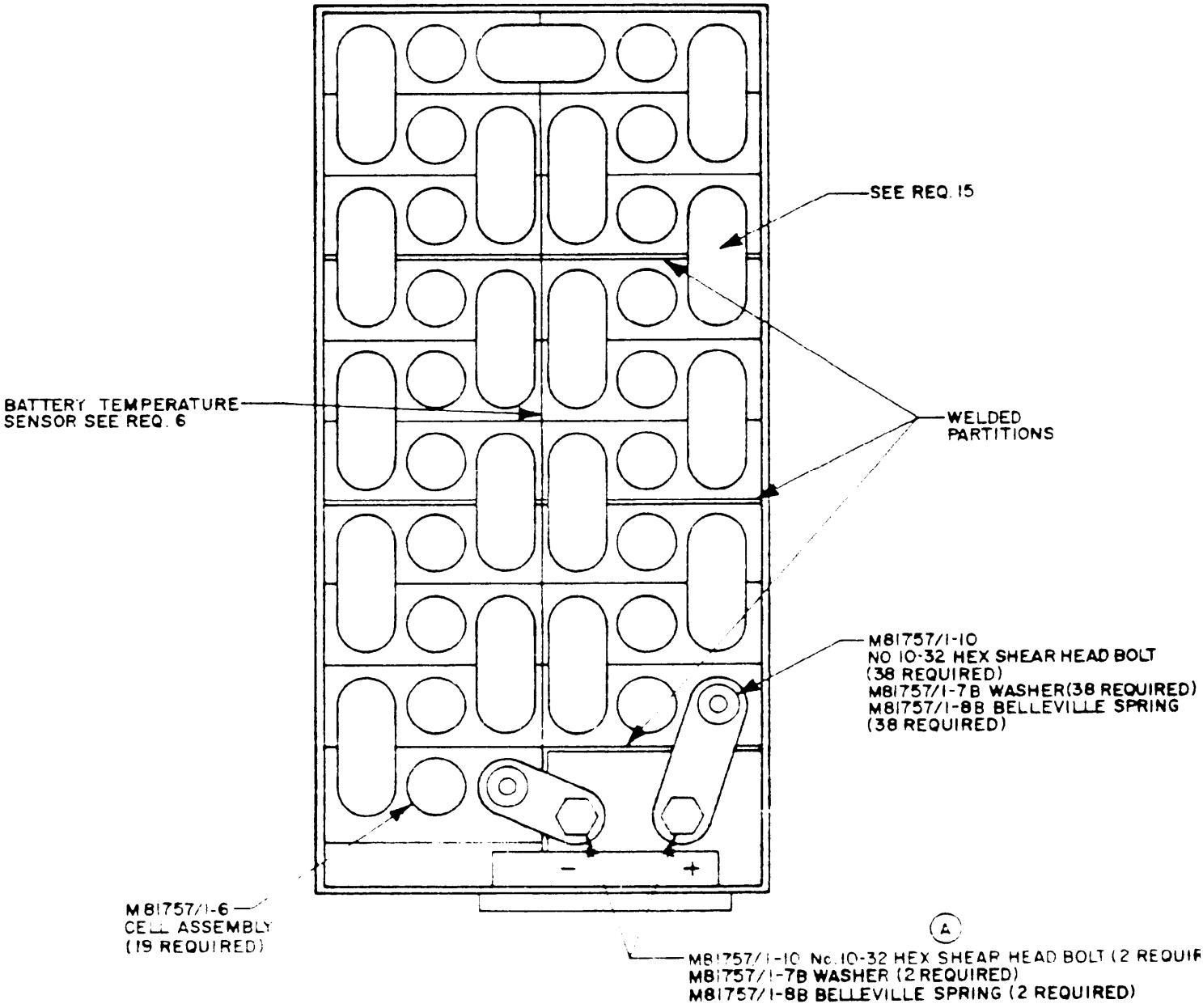


FIGURE 4. Cell arrangement.

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REQUIREMENTS:

1. The receptacle terminal block shall be 0.866 ± 0.015 from the centerline of the contact pins to the top surface of the terminal block. The receptacle shall be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MS3509-29 or -26.

2. All dimensions are prior to epoxy coating.

3. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Lock Division, American Hardware, Item Number 15834 or Nielson Latch, Number C-833314-2-SS-302; or equivalent.

4. Spacers shall be polyethylene sheet in accordance with ASTM-STD D4066-82, Composition A, Type I and Type III, minimum 0.015 inch thick, color natural.

5. Latches and strikes shall not be epoxy coated. The outer 0.875 inch on each end of the cover hold-down bar shall not be epoxy coated.

6. A sensor (thermocouple or thermistor), when required, shall be located as shown in Figure 4 for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and from side to side).

7. Battery potential shall be 24 volts (nominal).

8. Maximum battery weight shall be 80 pounds (with electrolyte).

9. The rated capacity at 1C rate shall be 30 A.H./1 Hr./80°F/18.0V.

10. Length and width of cover pad shall conform to the inside dimensions of the cover.

11. The length, width and thickness of cover pad projections shall provide sufficient pressure on the vented filler caps to hold the cells down during vibration.

12. The quantity and location of cover projection pads shall be determined by the vented filler caps.

13. Deviations from the basic configuration dimensions (14.578 x 6.500 nominal) shall be no lower than 2.06 inches from the base.

14. Type M81757/1-10 shear head bolts shall be used to secure the cell interconnecting straps for prevention of cell and receptacle removal.

15. The cell connector straps shall be constructed of copper conforming to Federal Specification QQ-C-502 and plated in accordance with Federal Specification QQ-N-290, Class 1, Grade A. Thickness of the straps shall be 0.1250 ± .010.

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16. The cell vented filler cap shall be in accordance with MIL-B-81757/2.

17. All cells within a battery shall be from the same production lot.

18. The battery shall conform to MIL-B-81757 except as modified by the following:

3.5.7 Cells. Delete.

3.5.7.1 Mountings. Delete.

3.6.5.2.1 Cell terminal marking. Delete last sentence.

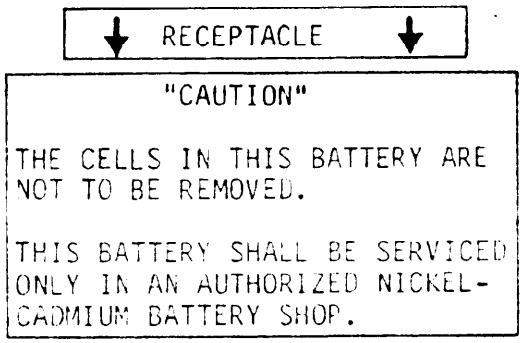
3.6.5.2.2 Cell identification. Delete.

3.6.5.2.3 Battery identification. Each battery shall have its identification marking silk screened with an epoxy overlay, or equal, that is impervious to 1.3 ± 0.025 specific gravity KOH electrolyte, in accordance with applicable specification sheet. Marking shall be in white print in accordance with MIL-M-13231. The marking shall show the following:

Information Required	Example
Battery, Storage	Battery, Storage
Military Battery Part No.	M81757/13-1
Replaces (include replacement data from applicable battery specification sheet)	Replaces MS18045-75, MS90321-75
National Stock Number	NSN 6140-01-***-****
Voltage (Nominal)	24 Volts
Maximum Weight	Maximum Weight 80 Lbs
Military "C" Rated Capacity	(30 AH/1 Hr/80°F/18.0V)
Contract Number	***
Date Code (see 3.6.5.2.5)	Date Code ***
Manufacturer's Part Number	***
(A) Manufacturer's Serial Number	***
Warranty Expiration Date	9-1/2

3.6.5.2.4 Battery caution and receptacle markings. These markings, in accordance with applicable specification sheet, shall be in white printing in accordance with MIL-M-13231. They shall be impervious to 1.3 ± 0.025 specific gravity KOH electrolyte. They shall be located as shown on the applicable specification sheet and read as follows:

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0.50 ± 0.03 Inch High Letters

0.50 ± 0.03 Inch High Letters

0.375 ± 0.03 Inch High Letters

0.375 ± 0.03 Inch High Letters

NOTES:

1. All dimensions are in inches.

2. Supersession data: This specification sheet supersedes MS18045 and MS90321 part numbers as follows:

Part Number	Replaces Part Number
M81757/13-1	MS18045-75 (Battery) MS90321-75W (Cell)

Preparing Activity:
Navy - AS
(Project No. 6140-N645)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-81757/13A(AS)	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM 24-VOLT, 20 AMPERE-HOUR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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